

Private should stay private.

Dirty COW • CVE-2016-5195

A local kernel race that changed protected file bytes
and enabled a fresh UID-0 login.

[COW](#) → [LIVE DUMMY DEMO](#) → [RECORDED ACCOUNT PROOF](#) → [DEFENCES](#)

Where private isolation breaks

The control and the exploit use different memory-write paths.

Normal COW control

O_RDONLY file

Writable MAP_PRIVATE mapping
memcpy changes a private copy

Backing file unchanged

Dirty COW race

O_RDONLY + read-only MAP_PRIVATE

Writer: pwrite to /proc/self/mem

Discard: MADV_DONTNEED

Vulnerable kernel → file changes

pwrites uses a virtual process address. The race is inside kernel COW/page handling.

Normal file permissions were already restrictive

Recorded environment: Ubuntu 12.04.2 • i686 / 32-bit • kernel 3.5.0-37-generic

seed UID 1000 | root:root 0644 target | seed-owned 0755 binaries | native ext4

```
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ id
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),109(lpadmin),124(sambashare),130(wireshark)
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ cat /zzz
11111122222333333
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ ls -l /zzz
-rw-r--r-- 1 root root 19 Oct  9 13:19 /zzz
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ sha256sum /zzz
7342e673ad4f9a5ea02fe84fa6ab1760d922e3f362c885ef4feda980423b9552 /zzz
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ echo 99999 > /zzz
bash: /zzz: Permission denied
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ echo $?
1
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ cat /zzz
11111122222333333
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$ sha256sum /zzz
7342e673ad4f9a5ea02fe84fa6ab1760d922e3f362c885ef4feda980423b9552 /zzz
[10/09/2026 13:25] seed@ubuntu:~/issd-member6/lab-files$
```

Actual S04 excerpt. Package: 3.5.0-37.58~precise1 / linux-lts-quantal. No Set-UID installation.

Switch to the live dummy demonstration

Prepare Part 0 first. Use the same guest terminal and a new run label.

1 Protection

id • permissions • original
file

Ordinary write should be
denied.

2 Normal COW

./cow_control

Private stars; backing file
original.

3 Bounded race

15-second classroom trial
Read and verify the whole
file.

```
bash run_trial.sh dummy 15 "$RUN"
```

Timing miss? Preserve it and show recorded S07: exact success, 0 integer seconds / 30-second bound.

Recorded file change → fresh UID-0 login

Task 2: only charlie's UID field changed. All other bytes and root:root 0644 were retained.

Original UID 1001 → field 0000 → fresh non-sudo login: UID 0 | GID stays 1002

```
[10/09/2026 14:22] seed@ubuntu:~/issd-member6/lab-files$ grep '^charlie:' /etc/passwd
charlie:x:0000:1002:,,,:/home/charlie:/bin/bash
[10/09/2026 14:22] seed@ubuntu:~/issd-member6/lab-files$ su - charlie
Password:
su: Authentication failure
[10/09/2026 14:25] seed@ubuntu:~/issd-member6/lab-files$ su - charlie
Password:
root@ubuntu:~# id
uid=0(root) gid=1002(charlie) groups=0(root),1002(charlie)
root@ubuntu:~# id -u
0
root@ubuntu:~# whoami
root
root@ubuntu:~# echo "Proof shell PID=$$"
Proof shell PID=3392
root@ubuntu:~#
```

S10 excerpt, 9 Oct 2026. First authentication failed; retry succeeded. Full original and recordings retained.

Fix the kernel; verify the restored state

All four required countermeasure themes. Optional patched comparison: not performed; discussion only.

Patch the kernel

Install the vendor fix and confirm the fixed kernel is actually booted.

Maintain updates

Use a supported OS and track package-family backports.

Limit local execution

Reduce unnecessary local access and untrusted code execution.

Monitor escalation

Check account integrity, numeric UID-0 entries and privileged sessions.

Verified afterward: old root shell ended • exact baseline • new UID 1001 login • cleanup

Recorded Task 1: the actual backing file changed

fresh-t1-01 • 30-second limit • 0 integer elapsed seconds • no finer runtime or attempt count measured

```
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ id
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),109(lpadmin),124(sambashare),130(wireshark)
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ pgrep -x cow_attack
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ diff -u logs/fresh-t1-01-before.txt logs/fresh-t1-01-after.txt
--- logs/fresh-t1-01-before.txt 2026-10-09 13:28:48.190096005 -0700
+++ logs/fresh-t1-01-after.txt 2026-10-09 13:28:48.414208000 -0700
@@ -1 +1 @@
-11111122222333333
+111111*****333333
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ cat /zzz
111111*****333333
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ ls -l /zzz
-rw-r--r-- 1 root root 19 Oct  9 13:19 /zzz
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ stat -c '%u:%g:%a:%s' /zzz
0:0:644:19
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ sha256sum /zzz
97c0ed6bb0deb556349015713d1c0a86d11e1bf5572a78b087650e814479b47b /zzz
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ cmp /zzz <(printf '111111*****333333\n')
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ echo $?
0
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ grep '^Elapsed:' logs/fresh-t1-01-summary.txt
Elapsed: 0s
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$
```

S07 readability excerpt: complete diff, fresh read, root:root 0644 / 19 bytes and exact-byte comparison.

Recorded Task 2: exact UID-only replacement

One ordinary-seed trial. 2040 bytes before and after; four-character UID field at byte offset 2002.

```
[10/09/2026 14:20] seed@ubuntu:~/issd-member6/lab-files$ grep '^charlie:' /etc/passwd
charlie:x:0000:1002:,,,:/home/charlie:/bin/bash
[10/09/2026 14:20] seed@ubuntu:~/issd-member6/lab-files$ diff -u logs/fresh-t2-01-before.txt logs/fresh-t2-01-after.txt
--- logs/fresh-t2-01-before.txt 2026-10-09 14:17:08.182233560 -0700
+++ logs/fresh-t2-01-after.txt 2026-10-09 14:17:08.406345561 -0700
@@ -39,4 +39,4 @@
telnetd:x:119:129:./nonexistent:/bin/false
vboxadd:x:999:1:./var/run/vboxadd:/bin/false
sshd:x:120:65534:./var/run/sshd:/usr/sbin/nologin
-charlie:x:1001:1002:,,,:/home/charlie:/bin/bash
+charlie:x:0000:1002:,,,:/home/charlie:/bin/bash
[10/09/2026 14:20] seed@ubuntu:~/issd-member6/lab-files$ stat -c '%u:%g:%a:%s' /etc/passwd
0:0:644:2040
[10/09/2026 14:20] seed@ubuntu:~/issd-member6/lab-files$ sha256sum /etc/passwd
99fced9d0853d999474f92617022e78efe7c87b0f63450da264c46bfa9d47cd2 /etc/passwd
[10/09/2026 14:20] seed@ubuntu:~/issd-member6/lab-files$ cmp /etc/passwd logs/fresh-t2-01-after.txt
[10/09/2026 14:21] seed@ubuntu:~/issd-member6/lab-files$ echo $?
0
[10/09/2026 14:21] seed@ubuntu:~/issd-member6/lab-files$ pgrep -x cow_attack
[10/09/2026 14:21] seed@ubuntu:~/issd-member6/lab-files$ grep '^Elapsed:' logs/fresh-t2-01-summary.txt
Elapsed: 0s
[10/09/2026 14:21] seed@ubuntu:~/issd-member6/lab-files$ awk -F: '$3 == 0 {print $1 ":" $3}' /etc/passwd
root:0
charlie:0000
[10/09/2026 14:21] seed@ubuntu:~/issd-member6/lab-files$
```

S09b: live file equals the saved after-copy; full-file verification separately confirms no other changes.

Restoration is a fresh-login check too

Proof shell PID 3392 was exited first. Exact normal bytes and a new ordinary login both passed.

```
[10/09/2026 14:29] seed@ubuntu:~/issd-member6/lab-files$ id
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),109(lpadmin),124(sambashare),130(wireshark)
[10/09/2026 14:29] seed@ubuntu:~/issd-member6/lab-files$ sudo -n cmp /etc/passwd /root/issd-member6-passwd.normal && echo
'Normal-account baseline restored'
Normal-account baseline restored
[10/09/2026 14:30] seed@ubuntu:~/issd-member6/lab-files$ sha256sum -c ../provenance/passwd-normal.sha256
/etc/passwd: OK
[10/09/2026 14:30] seed@ubuntu:~/issd-member6/lab-files$ grep '^charlie:' /etc/passwd
charlie:x:1001:1002:,,,:/home/charlie:/bin/bash
[10/09/2026 14:30] seed@ubuntu:~/issd-member6/lab-files$ id charlie
uid=1001(charlie) gid=1002(charlie) groups=1002(charlie)
[10/09/2026 14:30] seed@ubuntu:~/issd-member6/lab-files$ stat -c '%u:%g:%a:%s' /etc/passwd
0:0:644:2040
[10/09/2026 14:30] seed@ubuntu:~/issd-member6/lab-files$ su - charlie
Password:
charlie@ubuntu:~$ id
uid=1001(charlie) gid=1002(charlie) groups=1002(charlie)
charlie@ubuntu:~$ id -u
1001
charlie@ubuntu:~$ whoami
charlie
charlie@ubuntu:~$ exit
logout
[10/09/2026 14:31] seed@ubuntu:~/issd-member6/lab-files$ id
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),109(lpadmin),124(sambashare),130(wireshark)
[10/09/2026 14:31] seed@ubuntu:~/issd-member6/lab-files$
```

S11 excerpt. Existing process credentials survive file edits; close privileged shells before restoration.

Sources, evidence and assistance

Completed practical evidence is distinct from the future classroom demonstration.

Wenliang Du / SEED Labs — Dirty COW Attack Lab (2017)

seedsecuritylabs.org/Labs_20.04/Software/Dirty_COW/

Historical SEED12 manual • mmap(2) • madvise(2) • proc_pid_mem(5)

Ubuntu CVE-2016-5195 information • supplied Member 6 allocation

[REPORT.pdf](#) | [LIVE_DEMO.pdf](#) | [SLIDE_NOTES.pdf](#) | [video/TASK2_FALLBACK.mp4](#)

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